



Federal Board SSC-I Examination
Computer Science Model Question Paper
National Curriculum 2022-2023 (Inclusive Scheme of Studies 2024)
(50% MCQs & 50% Descriptive)

SECTION - A (Marks 30)

Time Allowed: 1:20 hours

Section – A is compulsory. All parts of this section are to be answered on the
OMR Sheet given in the E-Sheet and handed over to the Centre Superintendent.
Deleting/ overwriting is not allowed. Do not use lead pencil.

Roll Number							Paper Version
							A

Candidate Sign. _____

Invigilator Sign. _____

Note: Fill the correct bubble for each question on OMR Sheet given in the E-Sheet. Each question carries 0.5 mark.

S #	Question	(A)	(B)	(C)	(D)
1.	A computer model used in a weather forecasting system is primarily designed for:	To archive weather data permanently	To control weather monitoring equipment	To simulate atmospheric conditions and predict future weather	To calculate electricity consumption in weather stations
2.	What is the primary purpose of HTML in web development?	To design the layout and structure of a webpage	To add dynamic behavior and interactivity to a webpage	To manage databases on a website	To control computer operations
3.	Which tag is used to create a hyperlink in HTML?		<a>	<p>	<div>
4.	Which symbol is commonly used to end statements in JavaScript?	Colon (:)	Period (.)	Semi colon (;)	Comma (,)
5.	Why is visualization important in data analysis?	It increases file size	It makes data harder to read	It helps in understanding data easily	It removes unnecessary data
6.	A student organizes exam marks in a table to easily compare results. What does this represent?	Data collection	Data analysis	Data encryption	Data backup
7.	Which programming language is commonly used for web development	HTML	CSS	Java Script	Python
8.	A CSS rule is written as: </>CSS body{ background-color: yellow;} what will happen?	Text becomes yellow	Page background becomes yellow	Image turns yellow	Button color changes to yellow

9.	Students are requested to organize and show the relationships between different topics of Artificial Intelligence, such as machine learning, computer vision, and natural language processing. Which tool would be most suitable for this purpose?	Flowchart	Trace tables	Algorithm	Concept maps
10.	Which scenario best illustrates the transition from Process to Output in an IPO chart for a weather app?	Entering the city name and clicking search.	Calculating the average weekly temperature from raw data.	Converting calculated data into a visual 7-day forecast.	Refreshing the app to check for new server updates.
11.	In a voice assistant system, Natural Language Processing (NLP) is used to:	Convert audio signals into electrical power	Extract meaning and context from human speech input	Encrypt user voice data for security purposes	Connect the assistant to external hardware devices
12.	Which technology is used to build systems like chatbots that can interact with users and provide automated responses?	Machine learning	Expert system	Digital assistance	Computer vision
13.	A student tests a solution by executing it and fixing any errors that appear. Which stage of the problem-solving process is being carried out?	Decomposition	Debugging	Pattern recognition	Abstraction
14.	A student is unable to identify an error in a program. The teacher advises reviewing the flowchart created before coding. How can the flowchart help?	It will automatically correct the error	It will execute the program	It will show the logical sequence of steps and decisions	It will convert the program into machine code
15.	A student creates a web page and wants to display a notification popup. Which JavaScript function will achieve this?	alert()	print()	show()	message()
16.	A student is using several software applications at the same time and the computer becomes slow. Which component should be upgraded to improve performance?	ROM	RAM	Register	Cache
17.	While preparing for exams, a student studies previous question papers to identify similarities in question types and predict possible future questions. Which computational thinking concept is being used?	Decomposition	Pattern recognition	Abstraction	Algorithm

18.	Which of the following classifications represents categorical data?	Quantitative	Binary	Qualitative	Continuous
19.	Big data helps organization to:	Ignore Customers	Reduce Information	Avoid Technology	Make Better Decision
20.	A young entrepreneur evaluates a business idea by gathering customer feedback and studying competitors before making any financial investment. Which stage of the entrepreneurial process is being applied?	Production	Marketing	Empathize	Budgeting
21.	A developer wants to style a webpage by changing colors, fonts, and layout. Which language is used for this purpose?	CSS	HTML	ASP	Java script
22.	Which Technology is used in mobiles for Face recognition?	Computer Vision	NLP	Robotics	Machine learning
23.	Cloud computing primarily offers computing resources and services over the internet. What does it mainly provide?	Online storage and computing services	To replace all computer hardware	To provide internet browsing services only	Only data backup solutions
24.	A school installs the same licensed copy of Microsoft Windows on several computers using one activation key without buying separate licenses for each system. What type of software usage does this represent?	Freeware	Shareware	Pirated Software	Open Source software
25.	While using email, a student receives a message that appears to be from a bank and asks him to urgently enter his password and personal information via a link. What is this type of email known as?	Spam	Phishing	Cookie	Spyware
26.	How does the Control Unit manage instructions while a program is being executed?	It performs mathematical calculations and logic operations	It directs and coordinates all CPU activities during execution	It permanently stores user data and programs	It connects the computer to external hardware devices
27.	Based on the nature of its content, how can fake news be recognized?	True information	Misleading information	Educational content	Verified news
28.	What is the primary purpose of functions in JavaScript?	To store permanent data in the browser	To delete unwanted files from memory	To block advertisements on a webpage	To group reusable blocks of code for performing specific tasks

29.	Why does a computer use secondary storage along with primary memory?	To increase CPU speed	To store data permanently	To control input devices	To process instructions
30.	Which innovation has made it possible to conduct business and trade globally over the internet?	Digital marketing	E-Commerce	Social networking	Radio/ TV broadcasting
31.	Which ability helps entrepreneurs think of innovative and original business ideas?	Coding ability	Graphic designing	Creative thinking	Marketing analysis
32.	When a company collects and analyze users' private data without permission for profit, it leads to which issue?	Data optimization	Privacy violation	System efficiency	Digital collaboration
33.	A student spends most of the day on social media, resulting in poor academic performance and lack of sleep. Which harmful effect of computing does this reflect?	Enhanced productivity	Educational growth	Digital addiction	Improved health
34.	Why are registers regarded as the fastest type of memory in a computer system?	They store large amounts of data permanently	They are built directly inside the CPU for immediate access	They are used for long-term data backup	They are the least expensive type of memory
35.	How does an algorithm help in solving a problem effectively?	It organizes data in random order	It converts programs into machine code automatically	It deletes errors from a program automatically	It provides a clear step-by-step procedure to reach a solution
36.	What is the primary purpose of JavaScript in web development?	To structure the content of a webpage	To style the appearance of a webpage	To make webpages interactive and dynamic	To design the layout of a webpage
37.	Which topology connects all devices to one central device?	Bus	Ring	Star	Mesh
38.	In computer networking, what are protocols defined as?	Set of communication rules	Hardware design specifications	Storage devices used in computers	Programming languages used for software development
39.	Which technology enables everyday physical devices to connect to the internet and exchange data automatically?	Internet of Things	Cloud Computing	Embedded Systems	Device Networking
40.	The 3 Vs of big data are:	Volume, Variety, Velocity	Value, Validity, Visibility	Vision, Voice, Value	Volume, Verification, Vision

STIMULUS/SITUATION/SCENARIO BASED MCQs									
SCENARIO#1 QUESTIONS 41-44)		(FOR		A student is creating a simple website to share school announcements, mainly showing fixed content. To enhance interactivity, the student includes features like pop-up messages on clicks and voice-based controls using speech recognition. During development, the system is divided into modules such as short of attendance and exams schedule, while only key details are considered to keep the design simple. The student uses suitable software tools to write and edit the website code.					
41.	Which software tool can edit website code?	chrome	Visual Studio	Cassandra	MongoDB				
42.	During system development, the system is organized into separate modules such as attendance, examinations, and fees. This process is called:	Abstraction	Decomposition	Iteration	Debugging				
43.	An interaction using voice based controls into actions like saying “ Attendance status ” could open the attendance module is based on:	Computer vision	Speech Recognition (NLP)	Machine learning	Neural networks				
44.	How does modular design support updating a specific part of a system like the exam section?	By allowing changes in one module independently	It forces changes in the entire system	By merging all modules into one program	By converting code into machine language				
SCENARIO#2 QUESTIONS 45-48)		(FOR		A student is developing a JavaScript program to store and display exam marks using an array. The program creates an array of marks and accesses different elements using their index positions. <pre>let marks = [35, 50, 75, 100, 65]; document.write(marks[1] + "
"); document.write(marks[3] + "
"); document.write(marks.length);</pre>					
45.	What will be printed when the code executes?	35, 75, 5	50, 100, 5	50, 75, 4	35, 100, 4				
46.	Which JavaScript statement correctly adds the value 110 to the end of an array named marks?	marks.add(110)	marks.push(110)	marks.insert(110)	marks.append(110)				
47.	What does “ marks.length ” return in a JavaScript array?	Number of indexes starting from 0	Total number of elements in the array	Last index value of the array	Memory size of the array				
48.	Which JavaScript statement correctly calculates and displays the sum of the second and fourth elements of an array named marks?	document.write(marks[1] + marks[3]);	document.write(marks[2] + marks[4]);	document.write(marks.sum(1, 3));	document.write(marks.sum(2, 4));				
SCENARIO#3 QUESTIONS 49-52)		(FOR		A school is setting up a computer network for its lab. The technician connects all computers to a central device so that if one computer fails, the others keep working. However, the technician also considers another design where all computers are connected in a circular path, where failure of one node can affect the entire network. The goal is to choose a topology that is reliable, easy to manage, and suitable for future expansion.					

49.	The technician wants to reduce cable cost but accepts that a single cable failure may affect the whole network. Which topology fits this requirement?	Bus topology	Ring topology	Star topology	Bus & Ring topologies both				
50.	Why is the central device-based topology preferred in the school lab?	It uses less cable	It does not require any device	It is the cheapest	Failure of one node does not affect others				
51.	A technician is designing a school network using a central device where all computers are connected individually. One student claims: "If one computer fails in this setup, the entire network will stop working." What is the correct evaluation of this statement?	The statement is correct for star topology	The statement is correct for ring topology	The statement is incorrect for star topology	The statement is only correct for mesh topology				
52.	What is a key difference between star and ring topology in a school network?	Star topology has no connections between computers	Ring topology uses a central device	Star topology uses a central device for communication	Both have identical structures				
SCENARIO#4 (FOR QUESTIONS 53-56)		A data science team is working on a project to analyze student performance in schools. They collect data using online surveys, school databases, and sensors that track attendance automatically. The team ensures that the data is accurate, relevant, and properly organized before analysis. They also remove incomplete or duplicate entries to improve data quality.							
53.	When data is collected from users through Google Forms or online questionnaires, it is called:	Sensor-based collection method	Survey method	Simulation method	Experimental method				
54.	Why does the team remove duplicate and incomplete entries?	To increase storage usage	To improve data accuracy and quality	To reduce internet speed	To randomize the dataset structure				
55.	Which data collection method automatically collects information in real time without human involvement?	Surveys	Interviews	Sensors	Questionnaires				
56.	Which of the following is a primary data source in the scenario?	School databases	Published reports	Online surveys	Historical books				
SCENARIO#5 (FOR QUESTIONS 57-60)		A smart city uses interconnected digital systems to improve traffic management and public services. Sensors placed on roads collect real-time traffic data, which is analyzed to reduce congestion and optimize traffic signals. AI systems process this data to predict traffic flow and suggest alternate routes to drivers through mobile applications. Public transport systems are also monitored to improve efficiency and scheduling. The city stores large amounts of data in cloud systems for easy access and scalability. To ensure safety, cybersecurity measures and ethical data usage policies are implemented to protect citizens' information and maintain system reliability.							
57.	What is the main reason smart cities rely on cloud computing for data storage?	To limit data sharing between departments	To enable flexible storage and easy access	To reduce system functionality	To avoid storing large amounts of data				

			from anywhere						
58.	Why are cybersecurity measures important in a smart city system?	To increase traffic congestion	To limit the use of technology	To reduce internet connectivity	To ensure data privacy and prevent unauthorized access				
59.	What is the main purpose of installing sensors on roads in a smart city?	To store citizen personal data	To collect real-time traffic data	To replace traffic police completely	To provide internet to drivers				
60.	A driver receives AI-based suggestions for alternate routes during traffic congestion. What kind of action is the system performing?	To permanently store traffic records	Predicting and optimizing traffic flow in real time	To delete GPS data from vehicles	To increase fuel consumption				



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Time allowed: 1:40 hours

Total Marks: 30

Note: Answer all parts from Section „B“ and „C“ on the E-sheet at specifically provided space. Write your answers on the allotted /given spaces.

SECTION-B [Marks: 3x6 = 18]

1.	What is a business plan? Write down its two components.	3	OR	What is an operating system? Give any two examples.	3
2.	Compare simplex and Full Duplex Modes of communications with one example each.	3	OR	Differentiate between discrete and continuous data with one example of each.	3
3.	How can a user search for information on the internet in a safe and responsible way? (State any three methods)	3	OR	Compare simple and complex problems with one daily life example of each.	3
4.	A startup team struggles to design a mobile app and applies design thinking by understanding users' needs, generating ideas, and testing prototypes—how does this process help in creating innovative solutions? (Give any three points)	3	OR	A teacher uses graphs and charts to analyze student exam results in order to identify performance trends and make decisions. How can data visualization be applied in this situation? (Explain any two applications)	3
5.	Write down the purpose of any three input devices.	3	OR	List any three applications of Artificial Intelligence in real life.	3
6.	Identify and correct errors in the following JavaScript code: let fruits = ["Apple", "Banana", "Orange"]; console.log(fruits.length) console.log(fruit.length);	3	OR	Compare any two types of storage devices with respect to reliability, capacity and speed.	3

SECTION-C [Marks: 2x6 = 12]

7.	Explain how computing innovations can negatively affect society and the environment. Support your answer with real life examples.	6	OR	A school is setting up a computer network to connect its lab, library, and administrative office for efficient communication and data sharing. The network design follows the OSI model to ensure proper data transmission between devices. Explain how the OSI model helps in managing communication in this network by describing the role of any three layers.	6																																										
8.	Create a result sheet using HTML that can display the following data in table format. <table><tr><th>Roll No</th><th>Student Name</th><th>Math</th><th>English</th><th>Computer Science</th><th>Total Marks</th><th>Percentage</th></tr><tr><td>1</td><td>Ali Ahmed</td><td>78</td><td>85</td><td>90</td><td>253</td><td>84.3%</td></tr><tr><td>2</td><td>Sara Khan</td><td>88</td><td>79</td><td>92</td><td>259</td><td>86.3%</td></tr><tr><td>3</td><td>Usman Ali</td><td>65</td><td>72</td><td>70</td><td>207</td><td>69.0%</td></tr><tr><td>4</td><td>Ayesha Noor</td><td>90</td><td>88</td><td>95</td><td>273</td><td>91.0%</td></tr><tr><td>5</td><td>Hamza Raza</td><td>55</td><td>60</td><td>68</td><td>183</td><td>61.0%</td></tr></table>	Roll No	Student Name	Math	English	Computer Science	Total Marks	Percentage	1	Ali Ahmed	78	85	90	253	84.3%	2	Sara Khan	88	79	92	259	86.3%	3	Usman Ali	65	72	70	207	69.0%	4	Ayesha Noor	90	88	95	273	91.0%	5	Hamza Raza	55	60	68	183	61.0%	6	OR	Draw a flowchart that prints the sum of first 20 even numbers.	6
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